

L2.3 How Derivatives Affect the Shape of a Graph

Recall: L2.2 The Mean Value Theorem

Name _____

Solo – 9 minutes

1. Verify that the function satisfies the three hypotheses of Rolle's Theorem on the given interval. Then find all numbers c that satisfy the conclusion of Rolle's Theorem.

$$f(x) = x + \frac{1}{x} \quad \text{on } \left[\frac{1}{2}, 2\right]$$

Check the hypotheses one at a time, and only then use the conclusion. Say why any candidate you find is kept or rejected.